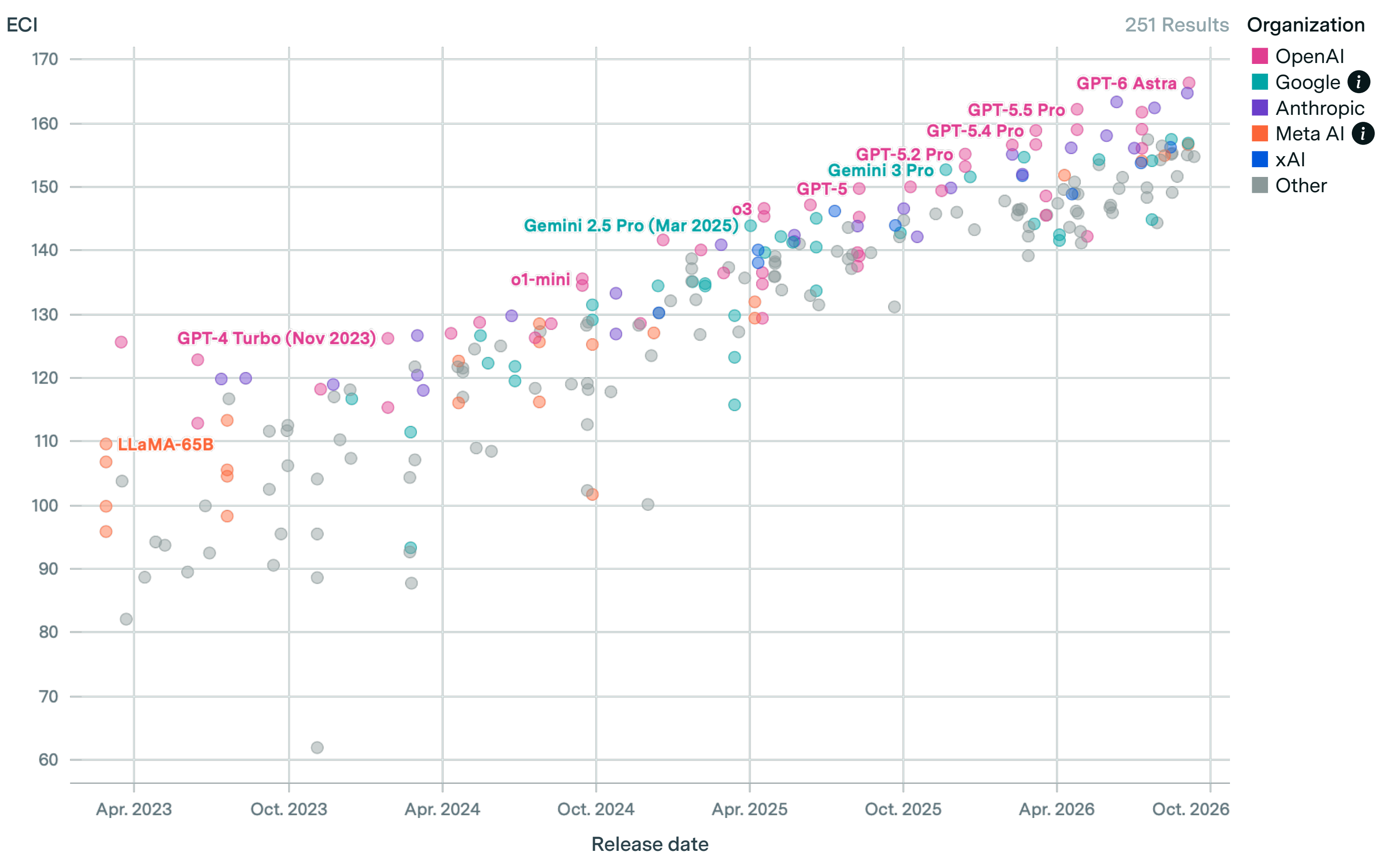
Epoch Capabilities Index

The Epoch Capabilities Index (ECI) combines scores from many different AI benchmarks into a single “general capability” scale, allowing comparisons between models even over timespans long enough for single benchmarks to reach saturation.

Learn more about ECI

Score vs time | Score vs training compute | Forest | Leaderboard | Domain vs general

Epoch Capabilities Index (ECI)



2023 ● 2026

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● General ECI
○ Software engineering ECI
○ Math ECI
○ Cyber ECI
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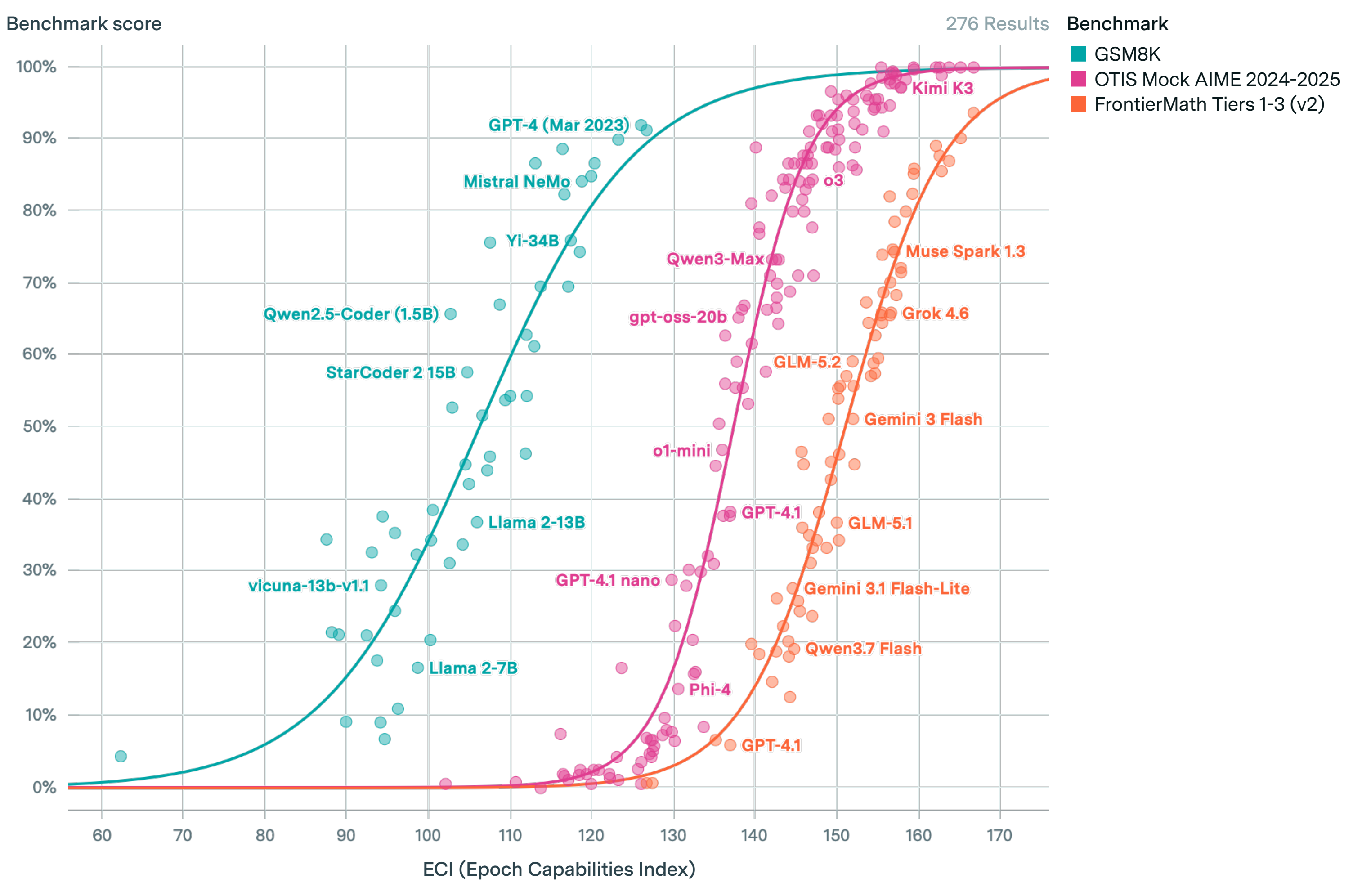
Select benchmarks +

Interpreting the ECI

An ECI value, on its own, is an abstract number. The chart below converts it back into concrete benchmark performance: for each benchmark, the curve shows the expected score a model with a given ECI would receive.

Each curve is a logistic fit to the actual data — every dot in the background is a real (ECI, score) observation we have for that benchmark. When the curve is steep, a little extra intelligence goes a long way, suggesting the benchmark will saturate quickly. Benchmarks with shallower, wider sigmoids are able to discriminate over a wider range of model capabilities. When scores are near 0% and 100%, benchmarks aren't very good at telling which model is more capable — a wide range of ECIs all correspond to a similar score.

From ECI to benchmark scores



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Search benchmarks...

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☐ Adversarial NLI
Model developer | Language understanding | 30–70%

☐ Aider Polyglot
Benchmark creator | Software engineering | Over 70%

☐ APEX-Agents
Benchmark creator | Agent | Over 70%

☐ ARC AI2
Model developer | Science, World knowledge | Over 70%

ARC-AGI-1

Common questions

What does the ECI represent? +

How should I interpret ECI values? +

Why isn’t this model’s ECI higher, if it leads some benchmarks? +

How do you decide which benchmarks to use? +

Why did the ECI score of a model change? +

Isn’t it a problem if model developers release only their best scores? +

Isn’t it a problem if model developers optimize for benchmark scores? +

Why isn’t my model included? +

Why isn’t my benchmark included? +

Why do some LLMs have a general ECI but not a Math ECI or SWE ECI? +

Why do some LLMs have a Cyber ECI but no general ECI? +

How should I interpret the results? e.g. What does it mean for an LLM to have a Math ECI of 160 but a general ECI of 155? +

Do the results mean that math and software engineering capabilities have been improving at the same rate as the general ECI? +

Did you add additional math or software engineering benchmarks to construct the SWE ECI or Math ECI? +

More about this dataset

Documentation
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Citations

Documentation

The general ECI is a composite metric which uses scores from over 50 distinct benchmarks to generate a single, general capability scale. At a high level, ECI stitches together its component benchmarks, determining their relative difficulty by making comparisons wherever models are evaluated on multiple benchmarks. Individual models obtain higher ECI scores if they perform better on harder benchmarks.

We give an overview of our methodology here; further technical details are available in our paper, A Rosetta Stone for AI Benchmarks, which was funded by Google DeepMind, and written in collaboration with researchers from their AGI Safety & Alignment team. However, the ECI is an independent Epoch AI product that Epoch has full rights over.

The Domain-specific ECIs show how LLMs’ ECI scores are influenced by the selection of benchmarks used to calculate them. It largely uses the methodology from the general ECI but only incorporates benchmarks from a specific domain, such as software engineering (SWE) or math. In particular, we keep the benchmark difficulty and slopes from the general ECI fit, and only refit the LLM capability parameters.

The Cyber ECI is constructed differently from other domain-specific ECIs, as it incorporates many cyber benchmarks that are not part of the general ECI. Due to this, it is updated less frequently than the general ECI. See our methodology for more details.

The results are given on a scale comparable to the general ECI, so if an LLM has a higher Math ECI than general ECI, this means it performs better on math benchmarks than non-math benchmarks. However this scaling means that the results cannot be used to assess long term progress trends in different domains, as they are all scaled to increase at the same overall pace as the general ECI (but it is possible to detect one-off capability jumps, or for e.g. one family of LLMs to go from underperforming to overperforming within a domain). See the methodology for the full details.

Code for the ECI is available in a public repository here.

Read the complete documentation

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